

(DUIS)

Consider we have funⁿ
 $f(x)$ with one const. say α .

$\therefore$ we'll have $f(x)$
 $\downarrow$
 $f(x, \alpha)$

$$I(a) = \int_2^{\infty} \frac{e^{-ax}}{x} dx$$

$f(x)$ $\frac{1}{x} e^{-ax} (-x)$

$\downarrow$

$f(x, a)$

To solve $\int_a^b f(x, \alpha) dx$

$$I(\alpha) = \int_a^b f(x, \alpha) dx$$

Using DUIS diff both side wrt. α

$$\frac{dI}{d\alpha} = \int_a^b \frac{\partial}{\partial \alpha} f(x, \alpha) dx$$

$$\frac{dI}{d\alpha} = h(\alpha)$$

$$\Rightarrow dI = h(\alpha) d\alpha$$
$$\int dI = \int h(\alpha) d\alpha + C$$
$$\Rightarrow I(\alpha) = \int h(\alpha) d\alpha + C$$

$$\int_0^{\infty} \frac{e^{-\alpha x} \sin x}{x} dx = \cot^{-1} \alpha \quad \text{Deduce}$$

$$\int_0^{\infty} \frac{\sin x}{x} dx = \frac{\pi}{2} //$$

$$I(\alpha) = \int_0^{\infty} \frac{e^{-\alpha x} \sin x}{x} dx$$

By DUIS

$$\frac{dI}{d\alpha} = \int_0^{\infty} \frac{\partial}{\partial \alpha} \left(\frac{e^{-\alpha x} \sin x}{x} \right) dx$$

$$\frac{dI}{d\alpha} = \int_0^{\infty} \frac{\sin x}{x} e^{-\alpha x} (-x) dx = - \int_0^{\infty} e^{-\alpha x} \sin x dx$$

$$\times \left[\begin{aligned} \int e^{ax} \cosh bx dx &= \frac{e^{ax}}{a^2 + b^2} (a \cosh bx + b \sinh bx) + c \\ \int e^{ax} \sinh bx dx &= \frac{e^{ax}}{a^2 + b^2} (a \sinh bx - b \cosh bx) + c \end{aligned} \right]$$

$$a = -\alpha, b = 1$$

$$\frac{dI}{d\alpha} = - \left[\frac{e^{-\alpha x}}{\alpha^2 + 1} (-\alpha \sin x - \cos x) \right]_0^{\infty} = - \left[0 - \frac{1}{\alpha^2 + 1} (0 - 1) \right]$$

$$= - \left[\frac{1}{\alpha^2 + 1} \right]$$

$$\int dI = \int - \frac{1}{(\alpha^2 + 1)} d\alpha + C \Rightarrow I(\alpha) = \cot^{-1} \alpha + C$$

$$\text{put } \alpha = \infty$$

$$I(\infty) = \cot^{-1}(\infty) + C$$

$$0 = 0 + C \Rightarrow C = 0$$

$$I(\alpha) = \int_0^{\infty} \frac{e^{-\alpha x} \sin x}{x} dx = \cot^{-1} \alpha$$

$$\text{put } \alpha = 0$$

$$\int_0^{\infty} \frac{\sin x}{x} dx = \cot^{-1}(0) = \frac{\pi}{2} //$$

$$\int_0^{\infty} \frac{e^{-x}}{x} (1 - e^{-ax}) dx$$

$$I(a) = \int_0^{\infty} \frac{e^{-x}}{x} (1 - e^{-ax}) dx$$

By DUIS.

$$\frac{dI}{da} = \int_0^{\infty} \frac{\partial}{\partial a} \left(\frac{e^{-x}}{x} (1 - e^{-ax}) \right) dx$$

$$= \int_0^{\infty} \frac{e^{-x}}{x} (0 - e^{-ax} (-x)) dx$$

$$= \int_0^{\infty} \frac{e^{-x}}{x} e^{-ax} dx = \int_0^{\infty} \frac{e^{-(1+a)x}}{x} dx$$

$$\frac{dI}{da} = \left[\frac{e^{-(1+a)x}}{-(1+a)} \right]_0^{\infty} = \left[0 - \frac{1}{-(1+a)} \right]$$

$$dI = \frac{1}{(1+a)} da$$

$$\int dI = \int \frac{da}{1+a} + C \Rightarrow I(a) = \log(1+a) + C$$

put $a=0$

$$I(0) = \log(1) + C$$

$$0 = 0 + C \Rightarrow C=0$$

$$I(a) = \log(1+a) //$$

$$\int_0^{\infty} \frac{\log(1+ax^2)}{x^2} dx = \pi\sqrt{a} \quad a \geq 0$$

$$I(a) = \int_0^{\infty} \frac{\log(1+ax^2)}{x^2} dx$$

$$\frac{dI}{da} = \int_0^{\infty} \frac{\partial}{\partial a} \left[\log \frac{(1+ax^2)}{x^2} \right] dx$$

$$= \int_0^{\infty} \frac{1}{\cancel{x^2}} \frac{1}{(1+ax^2)} (\cancel{x^2}) dx$$

$$\left. \int \frac{1}{x^2+a^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} \right\} = \int_0^{\infty} \frac{1}{a} \frac{1}{(\frac{1}{\sqrt{a}}+x^2)} dx$$

$$= \frac{1}{a} \int_0^{\infty} \frac{1}{(\frac{1}{\sqrt{a}})^2 + x^2} dx = \frac{1}{a} \left[\frac{1}{\sqrt{a}} \tan^{-1} \left(\frac{x}{\frac{1}{\sqrt{a}}} \right) \right]_0^{\infty}$$

$$= \frac{\sqrt{a}}{a} \left[\tan^{-1}(\sqrt{a}x) \right]_0^{\infty}$$

$$\frac{dI}{da} = \frac{\sqrt{a}}{a} \left[\frac{\pi}{2} - 0 \right] = \frac{1}{\sqrt{a}} \frac{\pi}{2}$$

$$dI = \frac{\pi}{2} (a)^{-1/2} da$$

$$\int dI = \frac{\pi}{2} \int a^{-1/2} da + C$$

$$I = \frac{\pi}{2} \frac{a^{1/2}}{1/2} + C \Rightarrow I(a) = \sqrt{a} \pi + C$$

$$\text{put } a=0 \Rightarrow I(0) = 0 + C$$

$$0 = 0 + C \Rightarrow C=0$$

$$I(a) = \sqrt{a} \pi$$

$$I(a) = \int_0^{\infty} \left(\frac{e^{-x} - e^{-ax}}{x \sec x} \right) dx$$

By DUs

$$\begin{aligned} \frac{dI}{da} &= \int_0^{\infty} \frac{\partial}{\partial a} \left(\frac{e^{-x} - e^{-ax}}{x \sec x} \right) dx \\ &= \int_0^{\infty} \frac{1}{x \sec x} (0 - e^{-ax} (-x)) dx \\ &= \int_0^{\infty} e^{-ax} \cos x dx \\ &= \left[\frac{e^{-ax}}{a^2 + 1} (-a \cos x + \sin x) \right]_0^{\infty} \\ &= \left[0 - \frac{1}{a^2 + 1} (-a) \right] = \frac{a}{a^2 + 1} \end{aligned}$$

$$\begin{aligned} \frac{dI}{da} = \frac{a}{a^2 + 1} &\Rightarrow \int dI = \int \frac{a}{a^2 + 1} da + C \\ I &= \frac{1}{2} \int \frac{2a}{a^2 + 1} da + C \\ I(a) &= \frac{1}{2} \log(a^2 + 1) + C \\ \text{put } a &= 1 \\ I(1) &= \frac{1}{2} \log(2) + C \\ 0 &= \log \sqrt{2} + C \Rightarrow C = -\log \sqrt{2} \\ I(a) &= \frac{1}{2} \log(a^2 + 1) - \log \sqrt{2} \\ &= \log \sqrt{\frac{a^2 + 1}{2}} \end{aligned}$$

p7. $\int_0^{\infty} e^{-(x^2 + \frac{a^2}{x^2})} dx = \frac{e^{-2a} \sqrt{\pi}}{2} //$

By DUs $I(a) = \int_0^{\infty} e^{-(x^2 + \frac{a^2}{x^2})} dx$

$$\begin{aligned} \frac{dI}{da} &= \int_0^{\infty} \frac{\partial}{\partial a} e^{-(x^2 + \frac{a^2}{x^2})} dx \\ &= \int_0^{\infty} e^{-(x^2 + \frac{a^2}{x^2})} \left(-\left(\frac{1}{x^2} 2a\right) \right) dx \end{aligned}$$

put $\frac{a}{x} = y \Rightarrow x = \frac{a}{y}$
 $-\frac{a}{x^2} dx = dy$

$x: 0 \rightarrow \infty \quad y: \infty \rightarrow 0$

$$\begin{aligned} &= 2 \int_{\infty}^0 e^{-(\frac{a^2}{y^2} + y^2)} dy \\ &= -2 \int_0^{\infty} e^{-(y^2 + \frac{a^2}{y^2})} dy = -2 I(a) \end{aligned}$$

$$\frac{dI}{da} = -2I \Rightarrow \frac{dI}{I} = -2 da$$

$$\Rightarrow \int \frac{dI}{I} = -2 \int da + \log C$$

$$\Rightarrow \log I = -2a + \log C \Rightarrow \log I = -2a + C$$

$$I = e^{-2a + C} = e^{-2a} e^C$$

$$I = e^{-2a} \Rightarrow I = C e^{-2a}$$

put $a = 0$

$$I(0) = C(1)$$

$$C = \int_0^{\infty} e^{-x^2} dx$$

$$x^2 = t$$

$$x = t^{1/2}$$

$$dx = \frac{1}{2} t^{-1/2} dt$$

$$x: 0 \rightarrow \infty \quad t: 0 \rightarrow \infty$$

$$C = \int_0^{\infty} e^{-t} \frac{1}{2} t^{-1/2} dt$$

$$= \frac{1}{2} \int_0^{\infty} e^{-t} t^{-1/2} dt = \frac{1}{2} \Gamma\left(\frac{1}{2}\right) = \frac{\sqrt{\pi}}{2}$$

$$\therefore I = \frac{\sqrt{\pi}}{2} e^{-2a}$$

$$\int_0^{\pi/2} \frac{\log(1+a \sin^2 x)}{\sin^2 x} dx = \pi [\sqrt{a+1} - 1]$$

$$I(a) = \int_0^{\pi/2} \log \frac{(1+a \sin^2 x)}{(\sin^2 x)} dx$$

by DUIS

$$\frac{dI}{da} = \int_0^{\pi/2} \frac{\partial}{\partial a} \log \frac{(1+a \sin^2 x)}{\sin^2 x} dx$$

$$= \int_0^{\pi/2} \frac{1}{\sin^2 x} \cdot \frac{1}{(1+a \sin^2 x)} \sin^2 x dx$$

Multiply & divide by $\frac{1}{\cos^2 x}$

$$= \int_0^{\pi/2} \frac{\frac{1}{\cos^2 x}}{\left(\frac{1}{\cos^2 x} + a \frac{\sin^2 x}{\cos^2 x}\right)} du = \int_0^{\pi/2} \frac{\sec^2 x}{(\sec^2 x + a \tan^2 x)} du$$

$$= \int_0^{\pi/2} \frac{\sec^2 x}{1 + \tan^2 x + a \tan^2 x} du$$

$$\begin{aligned} \tan x &= t \\ \sec^2 x dx &= dt \\ x: 0 \rightarrow \pi/2 &\quad t: 0 \rightarrow \infty \end{aligned}$$

$$= \int_0^{\infty} \frac{dt}{1 + t^2 + at^2}$$

$$= \int_0^{\infty} \frac{dt}{1 + (1+a)t^2} = \int_0^{\infty} \frac{1}{(1+a)} \cdot \frac{dt}{\left(\frac{1}{1+a} + t^2\right)}$$

$$\begin{aligned} &= \frac{1}{1+a} \int_0^{\infty} \frac{dt}{\left(\frac{1}{1+a}\right)^2 + t^2} = \frac{1}{1+a} \cdot \frac{1}{\sqrt{\frac{1}{1+a}}} \left[\tan^{-1} \left(\frac{t}{\sqrt{\frac{1}{1+a}}} \right) \right]_0^{\infty} \\ &= \frac{\sqrt{1+a}}{1+a} \left[\frac{\pi}{2} - 0 \right] \end{aligned}$$

$$\frac{dI}{da} = \frac{1}{\sqrt{1+a}} \cdot \frac{\pi}{2}$$

$$\int dI = \frac{\pi}{2} \int (1+a)^{-1/2} da + C \Rightarrow I = \frac{\pi}{2} \frac{(1+a)^{1/2}}{1/2} + C$$

Put $a = 0$

$$I(0) = \pi + C \Rightarrow 0 = \pi + C \Rightarrow C = -\pi$$

$$\begin{aligned} I &= \pi \sqrt{1+a} - \pi \\ &= \pi (\sqrt{1+a} - 1) \end{aligned}$$